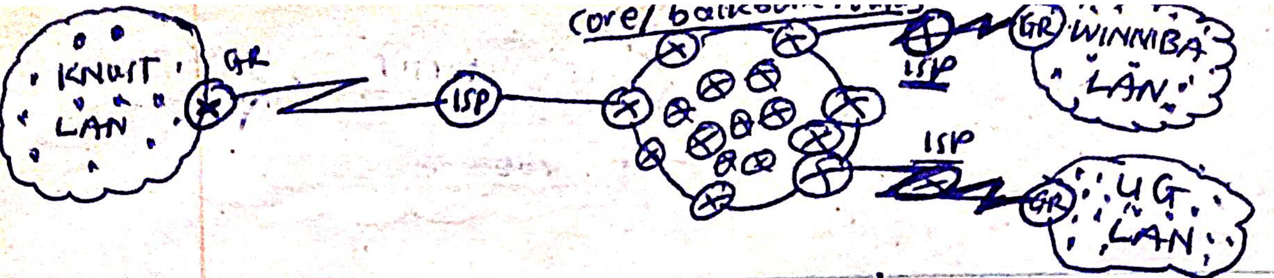


From the transport layer, datagrams or segments are passed to the network layer.

NETWORK LAYER.

IP

This is where routers and layer 3 switches operate. The data units received are termed as packets. At this layer, there is logical addressing whereby the IP assigns source and destination IP Addresses to the packets to make sure they arrive at their intended destination. IP Addresses used can be either IPv4 (32-bit) and IPv6 (128-bit).

✓ There is packet switching where TCP segments are tracked whiles UDP datagrams are not tracked.

✓ It uses IP address as the addressing mechanism.

IP ADDRESS.

✓ IPv4 [32 bits]

✓ IPv6 [128 bits]

IPv4

- 32 bits.

bit = 0, 1

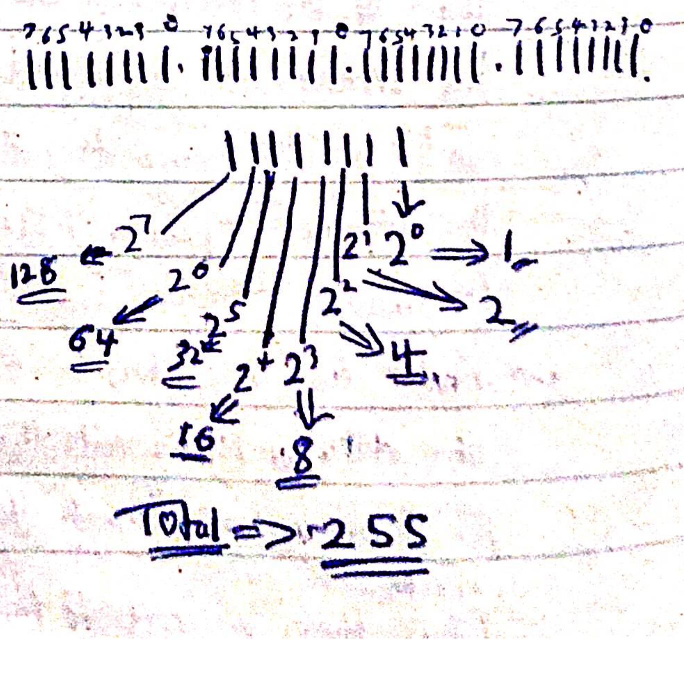
∴ 2^{32} = 4 billion.

NB: The world has a population of over 4 billion, so how is this addressing mechanism able to ~~cater for all~~ handle the IP addresses of the entire world.

Schemes Used

✓ IPv4 Classification

32 bits



Class A network ⇒ 0 1111111

start bit = 0,,
end bit = $255 - 2^7$
$= 255 - 128$
$= \underline{127}$

⇒ but since it can't start from 0 and 127 is also reserved
Class A ⇒ 1 — 126

IP Address

Network part + Host Part

N/B: A Network can't exist without a host

CLASS A

11111111 . 11111111 . 11111111 . 11111111

$\underbrace{11111111}_{N}$. $\underbrace{11111111}_{H}$. $\underbrace{11111111}_{H}$. $\underbrace{11111111}_{H}$

class A Network bits

7 6 5 4 3 2 1 0
1 1 1 1 1 1 1 1 /8

⇓

7 6 5 4 3 2 1 0
0 1 1 1 1 1 1 1

with 2^7 out, $2^7 = 128$.
Class A network = 255 − 128
= 127,,

That's, class A ⇒ 0 — 127.
but since a network can't start with 0, and class A has a reserved ~~bit~~ for LOOPBACK TESTING.
Class A starts from 1 — 126

LOOPBACK TESTING
Ping 127.0.0.1
test the OS for network connectivity

That is $2^7 = 128$ − [beginning and starting bit]

⇒ 128 − 2 = 126,,

∴ Class A has 126 valid networks

As said earlier, ~~network~~ 127.0.0.1 is reserved for LOOPBACK TESTING to test for network connectivity on the OS ⇒ Ping 127.0.0.1,,

CLASS A HOSTS

N. $\underset{H}{11111111}$. $\underset{H}{11111111}$. $\underset{H}{11111111}$

⇒ 24 bits

⇒ 2^{24} = 16,777,216 hosts,,

This is the number of class A hosts, but since there has to be a broadcast address for class A, 1 bit is reserved for that and another bit is reserved for network Identification.

∴ Valid Class A ~~network~~ hosts

⇒ 24 bits — 2 bits.

⇒ 2^{24} — 2

⇒ 16,777,214 Valid hosts

2 ⇒
- Broadcast Address
- Network Identification.

NB: Start bit = The number of 1's before the 0. (decimal)
end bit = 255 − the decimal value of the position of 0

Number of Networks ⇒ the decimal value of 1's after 0 assigned as network bits.

That means, each class A network has 2^{24} hosts and $(2^{24}-2)$ valid hosts.

⇒ 1.0.0.0 + 16,777,214 + BD address
⋮
126.0.0.0 + 16,777,214 + BD address

Identified Network | Valid Hosts | Broadcast Address

NB: This huge number of hosts on the network causes congestion on the network through multicast traffic, broadcast storm etc.
To solve this issue of congestion, we subnet the network. ⇒ LATER.

CLASS B

11111111.11111111.11111111.11111111
N N H H

Class B Networks

11111111.11111111 ⇒ /16

10111111.11111111

start bit for class B
$\Rightarrow 2^7 = 128$

The last bit for class B
$\Rightarrow 255 - 2^6$
(total) (10 end)

$\Rightarrow 255 - 64$
$\Rightarrow 191$

Number of Networks
10111111.11111111
14 bits.

∴ Class B has 2^{14} networks.
∴ Number of Valid networks.
$2^{14} - 2$ — Network Identification, Broadcast Address
$\Rightarrow 2^{14} - 2 = 16,382$

Number of Hosts
N.N.11111111.11111111
16 Host bits.

∴ No. of hosts = 2^{16}
No. of valid hosts = $2^{16} - 2$

Class B
128
|
191

CLASS C

11111111.11111111.11111111.11111111

N N N H

class C Networks

11111111.11111111.11111111 /24

⇓

$\overset{76543210}{11011111}$.11111111.11111111

starting value for class C

$1^7+1^6 \Rightarrow 2^7+2^6 = 128+64$

$\Rightarrow 192$

Ending value for class C

$255-0^5$

$\Rightarrow 255-2^5$

$\Rightarrow 255-32$

$\Rightarrow 223$

Number of Class C networks

$\underbrace{11011111.11111111.11111111}_{\text{21 bits.}}$

$2^{21} \Rightarrow$ total networks

Valid class C networks $= 2^{21}-2$

class C: 192 – 223

CLASS D

11111111.11111111.11111111.11111111

N N N N

⇓

$\overset{76543210}{11101111}$.11111111.11111111.11111111

starting value

$1^7+1^6+1^5 = 2^7+2^6+2^5$

$= 128+64+32$

$= 224$

Ending value

$255-0^4 = 255-2^4$

$\Rightarrow 255-16$

$\Rightarrow 239$

It's used for multicast transmission

class D: 224 – 239

2^{28} = Networks.

$2^{28}-2$ = Valid networks.

CLASS E

240 - 255

Used for Network researches and for experimentation.

The computer uses MASK and AND Gate to do it's network analysis.

⇒ There is a default MASK (boundary level MASK) which helps us to know ~~if~~ an IP address is subnetted or not subnetted.

NOTE [Defaults]

1) CLASS A network

11111111.00000000.00000000.00000000

default → Bitwise notation

default → Dotted notation

255.0.0.0

default CIDR notation

/8

with a /8 means the class A network has not been subnetted.

...../7 __ /9 .... /30

supernet network | subnet network

Its to /30 because 2 are not used.

NB: Every IP Address requires a MASK

We subnet networks to convert hosts to create networks.

All class A networks In Bitwise notation has

0111 _ _ . _ _ _ . _ _ _

start

last range

1 —— 126

126.24.3.24/16

/16 still means a class A network but its subnetted.

2) CLASS B Network

default Bitwise Notation.

11111111.11111111.00000000.00000000

default Dotted Notation

255.255.0.0

default CIDR notation

/16 ⇒ default for an unsubnetted class B.

....../15 /16 /17..... /30

supernet networks | default | subnet networks

All class B networks

bitwise notation

10 11.........

start end

Range = 128 — 191

eg. 129·22·4·13/8

/8 here still means that, the network is class B [129·] just that it is a supernet network.

3) CLASS C Network

default Bitwise Notation

11111111·11111111·11111111·00000000

default Dotted notation

255·255·255·0

default CIDR notation

/24 ⇒ default MASK

...... /23 —— /25 ... /30

Supernet network | Subnetted network

For all Class C networks

Bitwise notation has

1101.........

Start — Last range

Range = 192 —— 223

NB

CLASS A	1 – 126	/8
CLASS B	128 – 191	/16
CLASS C	192 – 223	/24
CLASS D	224 – 239	/30
CLASS E	240 – 255	

EXAMINATION TRICKS

①

IP Address ⇒ 187·132·65·7

Soln

1) Network Address: It's a class B network

∴ Network Address ⇒ 187·132·0·0

2) Network ID ⇒ 187·132

3) Host ID ⇒ 65·7

4) Last Valid host ⇒ 187·132·255·254 (host)

5) Broadcast Address ⇒ 187·132·255·255

②. 121.69.72.14

Soln

Its a class A network [121].

✓ Network Address -

121.0.0.0

✓ Network ID

121

✓ Host ID

69.72.14

✓ Last Valid Host address

~~121.69~~ → 121.255.255.254

✓ Broadcast address

⟹ 121.255.255.255//

FROM THIS DIAGRAM

IP Address

orbit-computer.solutions.com

hierarchical domain name

orbit-computer ⇓ 3rd level domain [the actual machine]

.solutions ⇓ 2nd level domain

.com ↓ top level domain ⇓ connects to the root server

Network Analysis of the Network diagram.

✓ 1 Network.

✓ 3 broadcast domains [host connected]

✓ 9 Collision domains.

✓ 3 segments [switches]

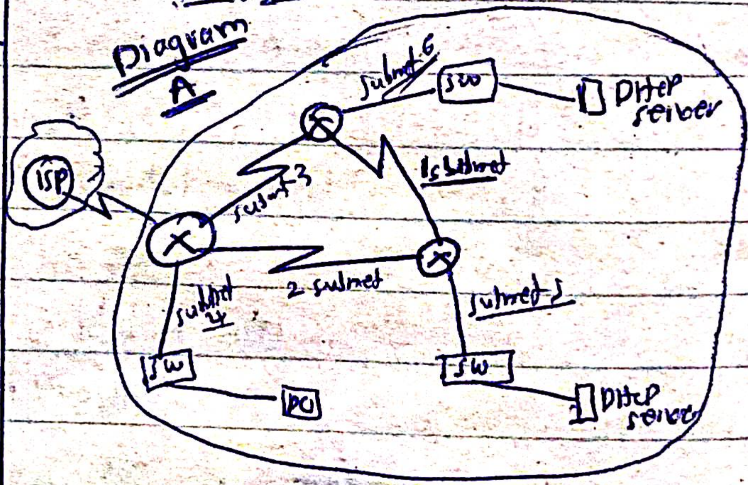
✓ 6 subnets [every router ~~interface~~ interface]

✓ It a client-server Architecture.

✓ A hybrid topology with a...?

6 Subnets

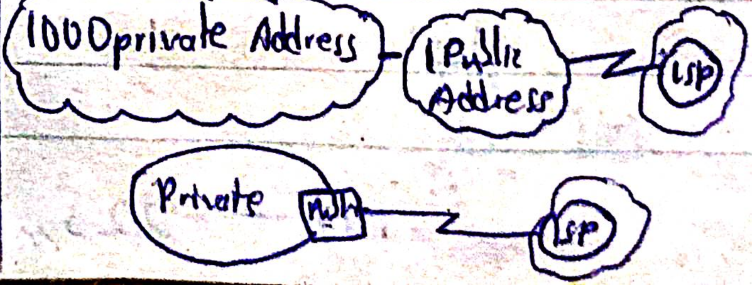
Diagram A



✓ 3 branches [segments]

Private Addressing

They are non-routeable addresses i.e not permited to be used on the internet.



2. 121·69·72·14

Soln

It's a class A network [121]

✓ Network Address –

121·0·0·0

✓ Network ID

121

✓ Host ID

69·72·14

✓ Last Valid Host address

~~121·69~~ 121·255·255·254

✓ Broadcast address

⟹ 121·255·255·255

FROM THIS DIAGRAM

IP Address

orbit-computer·solutions·com

hierarchical domain name

orbit-computer ⇓ 3rd level domain [the actual machine]

·solutions ⇓ 2nd level domain

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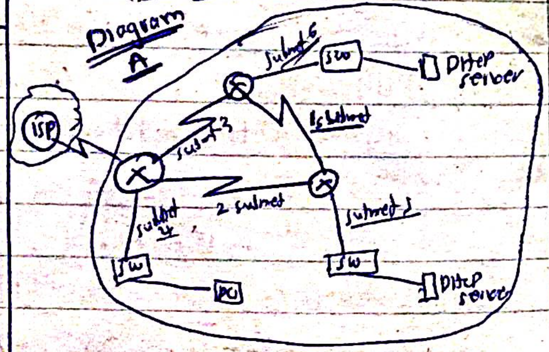
Diagram A



✓ 3 branches [segments]

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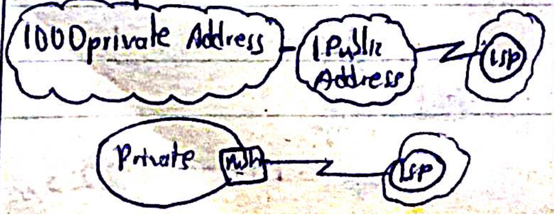
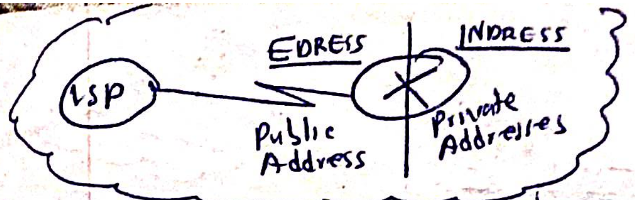




Reserved IP network numbers

CLASS A

10·0·0·0 ⟹ Network number.

Subnet Mask ⟹ 255·0·0·0

IP Address Range ⟹

10·0·0·1 — 10·255·255·254

CLASS B

Network Number: 172·16·0·0 – 172·31·0·0

Subnet Mask = 255·255·0·0

IP Address Range = 172·16·0·1 – 172·31·255·254

CLASS C

192·168·0·0

S·M ⟹ 255·255·255·0

IP·A·R ⟹ 192·168·0·1 – 192·168·255·254

⟹ Private addresses gets access to the internet through the NAT protocol. The NAT protocol maps all Private addresses to a public access to the internet.

Examinable.

Provide a network diagram using IP Address 192·168·16·5/24 for Diagram A.

Diagram Analysis

6 subnets.

3 ~~seg~~ segments.

3 BDs. each with 2 CDs.

9 CDs in all.

Using 192·168·16·5/24

✓ class C network

✓ Not subnetted ⟹ /24

Solution

We have to subnet the network to 6 subnets, [router interfaces].

Procedure

✓ FLSM = Fixed Length Subnet Mask

- This assigns equal number of hosts to each router interface.

✓ VLSM = Variable Length Subnet Mask

- This assigns variable number of hosts to each router interface to maximize the address usage.

FLSM

IP Address 192·168·16·5/24

Since it's a class C network

⟹ 192·168·16·0/24

It has 254 valid hosts,,

so, 192·168·16·5/24 is the public address for the network,,

$$\overbrace{\underbrace{192.168.16.}_{\text{Network ID}}\ \underbrace{5}_{\text{Host ID}}\ /\underbrace{24}_{\text{not subnetted}}}^{\text{IP Address}}$$

NOW: Subnetting a CLASS C network with FLSM

* Subnetting = converting hosts into networks

CLASS C network

default 11111111·11111111·11111111·$\underbrace{00000000}_{\text{Hosts.}}$

Subnetting = borrowing and converting host bits into network bits (0 → 1),,

7 6 5 4 3 2 1 0 block size

0 0 0 0 0 0 0 0

1 2 3 4 5 6 7 8

number of bits to be borrowed.

NB 1) 10000000

borrowing 1 bit

$2^1 = 2$,, ⟹ 2 subnets,,

block size = $2^7 = 128$,,

2) 11000000

borrowing 2 bits

$2^2 = 4$,,

∴ 4 subnets.

11000000 block size ⟹ $2^6 = 64$,,

Since we want 6 subnets, lets borrow more

3).

borrowing 3 bits

$2^3 = 8$

∴ 8 subnets,,

5 4 3 2 1 0

11100000

block size ⟹ $2^5 = 32$,,

MASK for the subnets

$\underbrace{11111111.11111111.11111111.11100000}_{\text{27 bits}}$

⟹ /27

∴ New network address

⟹ 192·168·16·0/27

Now you write the various subnets using the block size.

⟹ 32,,

7 6 5 4 3 2 1 0
11000000
2 bits
subnet = 2^2 = 4 subnets
blocksize = 2^6 = 64

NB: For FLSM

7 6 5 4 3 2 1 0
0 0 0 0 0 0 0 0 → The positions are used to find the block size

The number of bits converted are used to determine the subnets. eg 10000000
blocksize = 2^7, subnet = 2^1

The given network 192.168.16.5/24

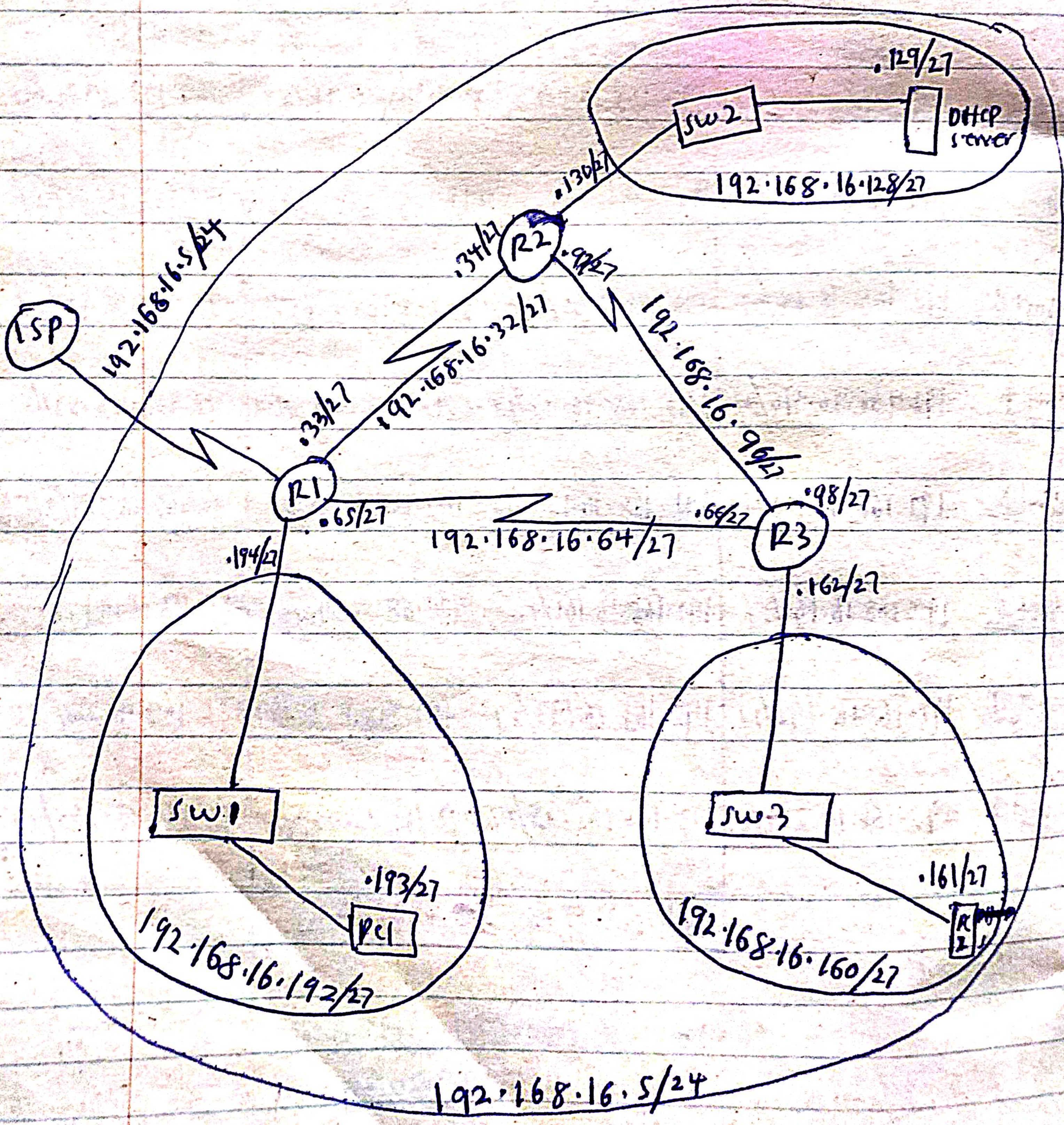
8 subnets

	Subnets	Ranges	Broadcast Addresses.
Subnet 1	192.168.16.0/27	192.168.16.1/27 — 192.168.16.30/27	192.168.16.31/27
subnet 2	192.168.16.32/27	192.168.16.33/27 — 192.168.16.62/27	192.168.16.63/27
subnet 3	192.168.16.64/27	192.168.16.65/27 — 192.168.16.94/27	192.168.16.95/27
subnet 4	192.168.16.96/27	192.168.16.97/27 — 192.168.16.126/27	192.168.96.127/27
subnet 5	192.168.16.128/27	192.168.16.129/27 — 192.168.16.158/27	192.168.16.159/27
subnet 6	192.168.16.160/27	192.168.16.161/27 — 192.168.16.190/27	192.168.16.191/27
subnet 7	192.168.16.192/27	192.168.16.193/27 — 192.168.16.222/27	192.168.16.223/27
subnet 8	192.168.16.224/27	192.168.16.225/27 192.168.16.254/27	192.168.16.255/27
		Each subnet has **30** valid hosts	

NOW WE PROCEED TO DESIGN THE NETWORK

FLSM

NETWORK DIAGRAM.



VLSM

The Diagram is subnetted but many IP Addresses are not being utilized since there are few hosts on each subnet:

eg: router interfaces only take 2 IP Addresses but the remaining 27 IP Addresses are wasted.

To ensure that IP Addresses are used/utilized effectively, we can use VLSM to assign the required IP Addresses for each subnet.

TWO SOLUTIONS

1) We can take one subnet of FLSM subnets and further subnet it with VLSM.

2) We can use VLSM by first Identifying the maximum number of Hosts required by each subnet in the network. We subnet it by considering the subnet with the highest maximum number of hosts.

[Here we have 2 hosts each so its recommended to fully use VLSM]

Solution 2

Fully Using VLSM.

VLSM focuses on the number of hosts on the subnet and assigns the required IP Addresses for each.

From the Diagram, we realize that each subnet requires a maximum of 2 hosts, and the subnet with the highest maximum number of hosts is also having 2 hosts.

Therefore, with highest maximum number of hosts being 2, we subnet the network with VLSM to get subnets with 2 hosts each in order to get 2 valid hosts each

SOLUTION

The given network 192.168.16.5/24

CLASS C

11111111.11111111.11111111.00000000

00000000

Total hosts = 2^7

Block size = 0 since no bit has been converted to 1

VLSM

hosts = number of 0s left

To get number of hosts, count the zeros from the right and get the decimal value, if its up to what we need, convert all the 0s on the left to 1s.

0 0 0 0 0 0 0 0
8 7 6 5 4 3 2 1

blocksize →

hosts ←

7 6 5 4 3 2 1 0
1 1 1 1 1 1 1 0
8 7 6 5 4 3 2 | 1 bit

Number of host = $2^1 = 2$

Blocksize = $2^1 = 2$ (not enough)

2)
7 6 5 4 3 2 1 0
1 1 1 1 1 1 0 0
8 7 6 5 4 3 | 2 1
2 bits.

Number of hosts = $2^2 = 4$

Number of valid hosts = $2^2 - 2 = 2$

Block size = $2^2 = 4$

Required

Now We proceed to get the new mask value.

11111111.11111111.11111111.11111100

/30

∴ The network for VLSM produced is

⟹ 192.168.16.0/30

with block size of 4

Number of Hosts ⟹ 4

Valid Number of Hosts ⟹ 2

Now we produce the subnets,

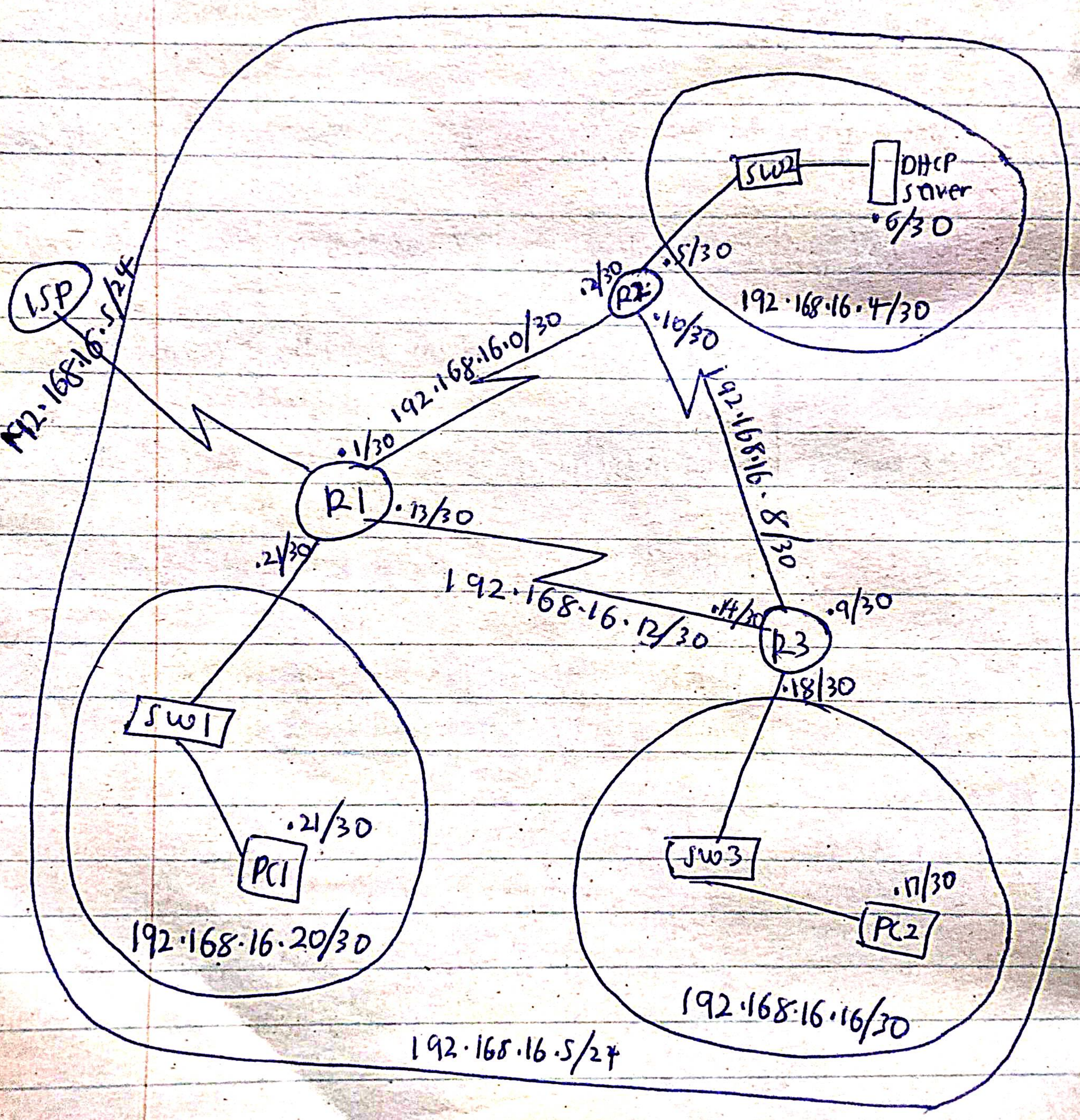
NB: since the network has 6 subnets, our table has just 6 subnets.

The given Network

⟹ 192.168.16.5/24

	Subnet	RANGES	Broadcast Address.
subnet 1	192.168.16.0/30	192.168.16.1/30 — 192.168.16.2/30	192.168.16.3/30
Subnet 2	192.168.16.4/30	192.168.16.5/30 — 192.168.16.6/30	192.168.16.7/30
Subnet 3	192.168.16.8/30	192.168.16.9/30 — 192.168.16.10/30	192.168.16.11/30
Subnet 4	192.168.16.12/30	192.168.16.13/30 — 192.168.16.14/30	192.168.16.15/30
Subnet 5	192.168.16.16/30	192.168.16.17/30 — 192.168.16.18/30	192.168.16.19/30
Subnet 6	192.168.16.20/30	192.168.16.21/30 — 192.168.16.22/30	192.168.16.23/30
	⋮		
	192.168.16.252/30	192.168.16.253/30 — 192.168.16.254/30	192.168.16.255/30
		Number of valid host for each subnet = 2	

NETWOR DIAGRAM



This Network ensures maximum and efficient utilization of IP Addresses such that no IP Address is wasted.

Solution 1 [FLSM → VLSM]

This occurs by taking 1 of the subnet of FLSM and breaking it down to get other subnets with VLSM.

⇒ 192.168.16.32/27.

range

192.168.16.63/27 ⇒ BC Address.

VLSM

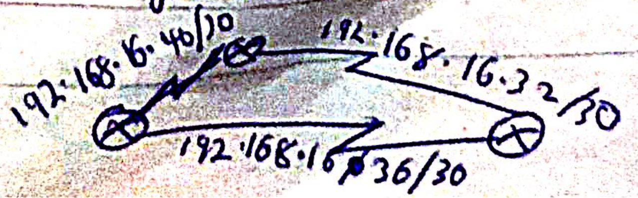
⇒ with blocksize = 4

MASK = /30

New Network	Range of valid hosts	Broadcast Address
192.168.16.32/30	192.168.16.33/30 – 192.168.16.34/30	192.168.16.35/30
192.168.16.36/30	192.168.16.37/30 – 192.168.16.38/30	192.168.16.39/30
192.168.16.40/30	192.168.16.41/30 – 192.168.16.42/30	192.168.16.43/30
⋮		
192.168.16.60/30	192.168.16.61/30 – 192.168.16.62/30	192.168.16.63/30

NB: 192.168.16.64/27 ⇒ is another subnet of FLSM

We use the new subnets for the network especially the backbone network



(Check the total number of subnets in the question before proceeding to answer)

FLSM PROCEDURE

Focus: Number of subnets

1) Determine the class in which the given network belongs to.

2) Write the default bitwise notation for the class of Network.

3). Write all the 0s and start converting these 0s which are hosts to get the ~~per~~ subnets.

4). Finding the blocksize, MASK and subnet.

SUBNET

Each 0 is converted to 1 from the left and upon every conversion check the decimal value of the number of 0s converted to 1s.

eg1) 11000000

This has 2 0s converted to ~~one~~ 1s.

∴ The subnet is $2^2 = 4$

eg 2) 11110000

Subnet $= 2^4 = 16$

16, 2^4, 25

BLOCKSIZE

The ~~bit~~ position of the last 1 bit gives the block size.

eg. 11000000 (positions 7 6 5 4 3 2 1 0) position 6

blocksize $= 2^6 = 64$

eg2) 11110000 (positions 7 6 5 4 3 2 1 0)

blocksize $= 2^4 = 16$

MASK

Rewrite the default network and add the new host bitwise notation.

1) 11111111 - 11111111 - 11111111 - 11000000

/26

2) 11111111 · 11111111 · 11111111 · 11110000

/28

5) Rewrite the network with the new network address.

6). Produce a range of all the networks with a range of their respective valid hosts and their broadcast addresses.

7) Use that to subnet the network.

check all the host required by each subnet and use the host value of the subnet with the highest number of hosts to subnet

1
96
16
112
16
128

VLSM PROCEDURE

Focus: Number of Hosts

1) Determine the class in which the given network belongs to

⋮

3) — — — —

4). finding the blocksize, MASK and SUBNET

SUBNET

find the number of host required by counting the 0s from the right.

eg. 1111111 0 ← 1 bit

$2^1 = 2$ hosts

eg2) 11111100 (2 bits)

$= 2^2 = 4$ hosts

eg3) 11111000 (3 bits)

$\Rightarrow 2^3 = 8$ hosts.

use this to get the maximum number of host you require.

eg if answer => 11111000 (3)

$2^3 = 8$ hosts

BLOCKSIZE

same as FLSM procedure

11111000 (3 2 1 0)

$2^3 = 8$

MASK

same as FLSM procedure

$\underbrace{11111111.11111111.11111111.11111}_{/29}000$

5) — — — —

7) — — — — — —